Hongchi Xia

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Research Interests

Spatial Intelligence for Embodied Physical AI, especially 3D Scene Reconstruction, Understanding, and Generation, as well as Simulation with Interactive Digital Twins for Embodied AI.

Education

University of Illinois Urbana-Champaign — Aug. 2024–present
Ph.D. in Computer Science; Advisor: Prof. Shenlong Wang — *Champaign, IL*

Shanghai Jiao Tong University — Sep. 2020–Jun. 2024
B.Eng. in Computer Science and Technology; GPA: 4.07/4.3; Rank: 1/106 — *Shanghai, China*

Publications

* denotes equal contribution. My name is bolded.

[1] **Hongchi Xia**, Xuan Li, Zhaoshuo Li, Qianli Ma, Jiashu Xu, Ming-Yu Liu, Yin Cui, Tsung-Yi Lin, Wei-Chiu Ma, Shenlong Wang, Shuran Song, and Fangyin Wei. "SAGE: Scalable Agentic 3D Scene Generation for Embodied AI." *CVPR, 2026*.

[2] **Hongchi Xia**, Chih-Hao Lin, Hao-Yu Hsu, Quentin Leboutet, Katelyn Gao, Michael Paulitsch, Benjamin Ummenhofer, and Shenlong Wang. "HoloScene: Simulation-Ready Interactive 3D Worlds from a Single Video." *NeurIPS, 2025*.

[3] **Hongchi Xia**, Entong Su, Marius Memmel, Arhan Jain, Raymond Yu, Numfor Mbiziwo-Tiapo, Ali Farhadi, Abhishek Gupta, Shenlong Wang, and Wei-Chiu Ma. "DRAWER: Digital Reconstruction and Articulation With Environment Realism." *CVPR, 2025*.

[4] Hao-Yu Hsu, Chih-Hao Lin, Albert J. Zhai, **Hongchi Xia**, and Shenlong Wang. "AutoVFX: Physically Realistic Video Editing from Natural Language Instructions." *3DV, 2025*.

[5] **Hongchi Xia**, Zhi-Hao Lin, Wei-Chiu Ma, and Shenlong Wang. "Video2Game: Real-time, Interactive, Realistic and Browser-Compatible Environment from a Single Video." *CVPR, 2024*.

[6] **Hongchi Xia***, Yang Fu*, Sifei Liu, and Xiaolong Wang. "RGBD Objects in the Wild: Scaling Real-World 3D Object Learning from RGB-D Video." *CVPR, 2024*.

[7] Bo Pang*, **Hongchi Xia***, and Cewu Lu. "Unsupervised 3D Point Cloud Representation Learning by Triangle-Constrained Contrast for Autonomous Driving." *CVPR, 2023*.

Selected Projects and Open Research Releases

- **Cosmos 3: Omnimodal World Models for Physical AI** — *NVIDIA Cosmos Lab, 2026*
  Research contributor to an open family of omnimodal world models for jointly processing and generating language, images, video, audio, and action for physical AI. [project page]
- **OpenPI Comet: Competition Solution for the 2025 BEHAVIOR Challenge** — *2025*
  Project contributor to a foundation-model adaptation and training system for long-horizon mobile manipulation in the BEHAVIOR Challenge. [paper]

Research Experience

Computer Vision Group, University of Illinois Urbana-Champaign — Jan. 2023–present
Researcher; Advisors: Prof. Shenlong Wang and Prof. Wei-Chiu Ma — *Champaign, IL*

- Developed HoloScene, a framework for reconstructing complete, physically stable, simulation-ready interactive worlds from a single video.
- Developed DRAWER, an automatic framework for perceiving, reconstructing, articulating, and re-simulating interactive objects in realistic 3D scenes.
- Developed Video2Game, which converts casually captured videos into real-time, realistic, browser-compatible game environments and robot simulators.

NVIDIA Cosmos Lab — May 2026–present
Research Intern; Manager: Ming-Yu Liu — *Santa Clara, CA*

- Contributed to Cosmos 3, an open family of omnimodal world models that jointly processes and generates language, images, video, audio, and action for physical AI.

Deep Imagination Research, NVIDIA May 2025–Feb. 2026
Research Intern; Mentor: Fangyin Wei; Manager: Ming-Yu Liu *Santa Clara, CA*

- Developed SAGE, a scalable agentic framework that generates simulation-ready 3D scenes from user specifications and supports downstream embodied action generation; released the SAGE-10K dataset.
- Contributed to OpenPI Comet, a foundation-model adaptation and training solution for long-horizon mobile manipulation in the 2025 BEHAVIOR Challenge.

Artificial Intelligence Group, University of California San Diego Jun. 2023–Feb. 2024
Visiting Researcher; Advisor: Prof. Xiaolong Wang *San Diego, CA*

- Created the WildRGB-D dataset, comprising approximately 8,500 real-world objects and nearly 20,000 RGB-D videos across 46 categories.

Machine Vision and Intelligence Group, Shanghai Jiao Tong University Aug. 2022–Jan. 2023
Research Assistant; Advisor: Prof. Cewu Lu *Shanghai, China*

- Developed Triangle-Constrained Contrast for unsupervised 3D point-cloud representation learning in autonomous-driving scenes.